How to Use LM317 for Making a Variable Power Supply Circuit

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Last Updated on November 10, 2024 by Swagatam (https://www.homemade-circuits.com/about/)

In this post I will elaborately explain how to build a simple LM317 based adjustable power supply circuit using minimum number of external components.

Table of Contents

- Introduction
- Essential Features
- How to Use LM317 for Producing a Variable Output
- How to Connect LM317 in a Adjustable Voltage Power Supply Design
- Output Voltage Formula
- Current Through Resistor R1
- Adjusting the Output Voltage
- Maximum Output Voltage
- Power Dissipation in the LM317
- Adding a Current Control Feature
-
- Enhancing with High Current Output Facility
- LM317 Variable Voltage, Current Regulator
- Circuit Operation
- Calculating the Output Voltage
- Calculating Current Limit
- Upgrading LM317 with Push Button Voltage Control
- High Current LM317 Bench Power Supply

As the name suggests a variable power supply circuit provides the user with a range of linearly varying output voltages through a manually controlled potentiometer rotation.

A LM317 is a versatile device which helps an electronic hobbyist to build a variable voltage power supply quickly, cheaply and very efficiently.

Introduction

Whether it's an electronic noob or an expert professional, an adjustable power supply (https://www.homemade-circuits.com/category/ac-dc-adapter/) unit is required by everybody in the field.

It is the basic source of power that may be required for various electronic procedures, right from powering intricate electronic circuits to the robust electromechanical devices like motors, relays etc.

A variable power supply unit (https://www.homemade-circuits.com/2013/06/simplest-variable-bench-power-supply.html) is a must for every electrical and electronic work bench and it's available in a variety of shapes and sizes in the market and also in the form of schematics to us.
These may be built using discrete components like transistors, resistors etc. or incorporating a single chip for the active functions.

No matter what the type may be, a power supply unit should incorporate the following features to become a universal and reliable with its nature:

Essential Features

- It should be fully and continuously adjustable with its voltage and current outputs.
- Variable current feature can be taken as an optional feature because it's not an absolute requirement with a power

supply, unless the usage is in the range of critical evaluations.
- The voltage produced should be perfectly regulated.

With the advent of chips or ICs like LM317, L200, LM338 (https://www.homemade-circuits.com/2012/04/ic-lm338-application-circuits-explained.html), LM723, configuring power supply circuits with variable voltage output with the above exceptional qualities has become very easy nowadays.

How to Use LM317 for Producing a Variable Output

Here I have explained how to construct a simplest power supply circuit (https://www.homemade-circuits.com/2012/03/how-to-design-power-supply-simplest-to.html) using the IC LM317. This IC is normally available in TO-220 package and has three pin outs.

The pin outs are very easy to understand, since it consists of an input, an output and an adjustment pins that just needs to be wired up with the relevant connections.

The input pin is applied with a rectified DC input, preferably with the maximum tolerable input, that's 24 volts as per the specs of the IC.

The output is received from the "out" pin of the IC while the voltage setting components are connected around the adjustment pin.

How to Connect LM317 in a Adjustable Voltage Power Supply Design

As can be seen the diagram, the assembly needs hardly any components and is in fact a child's play to get everything in place.

Adjusting the pot produces a linearly varying voltage at the output that may be right from 1.25 volts to the maximum level supplied at the input of the Ic.

Though the shown design is the simplest one and therefore includes only a voltage control feature, a current control feature can also be included with the IC.

Output Voltage Formula

The output voltage (V_{out}) of the LM317 can be calculated through the two external resistors, R_1 and R_2, using the formula:

$V_{out} = 1.25 * (1 + R_2 / R_1)$

Where:

- 1.25V is the reference voltage between the Adjust and Output pins.
- R1 is the resistor between the Output pin and the Adjust pin (typically 240Ω).
- R2 is the resistor between the Adjust pin and ground.

Current Through Resistor R1

The current I1 through resistor R1 is approximately:

$I_1 = 1.25 / R_1$

If $R_1 = 240\Omega$, the current is:

$I_1 \approx 5.21$ mA

Adjusting the Output Voltage

To set a desired output voltage we can choose the resistors R1 and R2 accordingly. For example if R1 = 240Ω and you want V_{out} = 5V then you can rearrange the formula for V_{out} to solve for R2:

$5V = 1.25V * (1 + R2 / 240\Omega)$

Solving for R2 gives:

$R2 = 720\Omega$

Maximum Output Voltage

The output voltage is limited by the input voltage and the LM317's dropout voltage, which is typically around 3V. The maximum output voltage (V_out) is:

$V_{out} = V_{in} - V_{dropout}$

For example if V_{in} = 12V and the dropout voltage is 3V then maximum output voltage would be:

$V_{out} = 12V - 3V = 9V$

Power Dissipation in the LM317

The LM317 dissipates power as heat, and this power dissipation (P_{diss}) is calculated as:

$P_{diss} = (V_{in} - V_{out}) * I_{out}$

Where:

- V_{in} is the input voltage.

- V_{out} is the output voltage.
- I_{out} is the output current.

For example if V_{in} = 12V, V_{out} = 5V, and I_{out} = 1A, then power dissipated by the LM317 would be:

P_{diss} = (12V - 5V) * 1A = 7W

Adding a Current Control Feature

The figure above shows, how the IC LM317 can be effectively used for producing variable voltages and currents, as desired by the user.

The 5K pot is used for adjusting the voltage, whereas the 1 Ohm current sensing resistor is selected appropriately to acquire the desired current limit.

Enhancing with High Current Output Facility

The IC can be further enhanced for producing currents higher than its rated values. The diagram below shows how the IC 317 can be used for producing more than 3 amps of current.

LM317 Variable Voltage, Current Regulator

Our versatile IC LM317/338/396 may be used as an adjustable voltage and current regulator through simple configurations.

The idea was built and tested by one of the avid readers of this blog Mr. Steven Chiverton and used for driving special laser diodes which are known to have stringent operating specifications, and could be driven only through specialized driver circuits.

The discussed LM317 configuration is so accurate that it becomes ideally suitable for all such specialist current and voltage regulated applications.

Circuit Operation

Referring to the shown circuit diagram, the configuration looks pretty straightforward, two LM317 IC s can be seen, one configured in its standard voltage regulator mode and the other in a current control mode

configured in its standard voltage regulator mode and the other in a current control mode.

To be precise the upper LM317 forms the current regulator stage while the lower acts like a voltage controller stage.

The input supply source is connected across the Vin and ground of the upper current regulator circuit, the output from this stage goes to the input of the lower LM317 variable voltage regulator stage.

Basically both the stages are connected in series for implementing a complete foolproof voltage and current regulation for the connected load which is a laser diode in the present case.

R2 is selected to acquire a range of around 1.25A max current limit, the minimum allowable being 5mA when the full 250 ohms is set in the path, meaning the current to the laser may be set as desired, anywhere between 5mA to 1 amp.

Calculating the Output Voltage

The output voltage of a LM317 power supply circuit could be determined with the following formula:

$$V_O = V_{REF} (1 + R_2 / R_1) + (I_{ADJ} \times R_2)$$

where is = V_{REF} = 1.25

Current ADJ is usually around 50 µA and therefore too negligible in most applications. You can ignore this.

Calculating Current Limit

The above is calculated by using the following formula:

R = 1.25/max allowable current

The current controlled voltage acquired from the upper stage is next applied to the lower LM317 voltage regulator circuit, which enables the desired voltage to be set anywhere from 1.25V to 30V, here the max range being 9V since the source is a 9V battery. This is achieved by adjusting R4.

The discussed circuit is assigned to handle not more than 1.5amps, if higher current is required, both the ICs may be replaced with LM338 for obtaining a max 5amp current or LM396 for a max of 10amp current.

The following lovely pictures were sent by Mr. Steven Chiverton, after the circuit was built and verified successfully by him.

Prototype Images

(https://www.homemade-circuits.com/wp-content/uploads/2014/08/current-controll-voltage-controll-driver-for-laser-diodes-1.jpg)

(https://www.homemade-circuits.com/wp-content/uploads/2014/08/2nd-laser-driver-power-supply-001-1.jpg)

(https://www.homemade-circuits.com/wp-content/uploads/2014/08/2nd-laser-driver-power-supply-002-1.jpg)

(https://www.homemade-circuits.com/wp-content/uploads/2014/08/2nd-laser-driver-power-supply-003-1.jpg)

(https://www.homemade-circuits.com/wp-content/uploads/2014/08/2nd-laser-driver-power-supply-004-1.jpg)

(https://www.homemade-circuits.com/wp-content/uploads/2014/08/2nd-laser-driver-power-supply-005-1.jpg)

(https://www.homemade-circuits.com/wp-content/uploads/2014/08/2nd-laser-driver-power-supply-009-1.jpg)

Upgrading LM317 with Push Button Voltage Control

So far we have learned how to configure an LM317 for producing adjustable output using a pot, now I have explained how push buttons may be used for enabling digitally controlled voltage selection.

We eliminate the use of mechanical pot and replace it with a couple of push buttons for the up/down selection of the desired voltage levels.

The innovation converts the traditional LM317 power supply design into a digital power supply design, by eliminating the low tech potentiometer which might be prone to wear and tear in the long run resulting in erratic operations and incorrect voltage outputs.

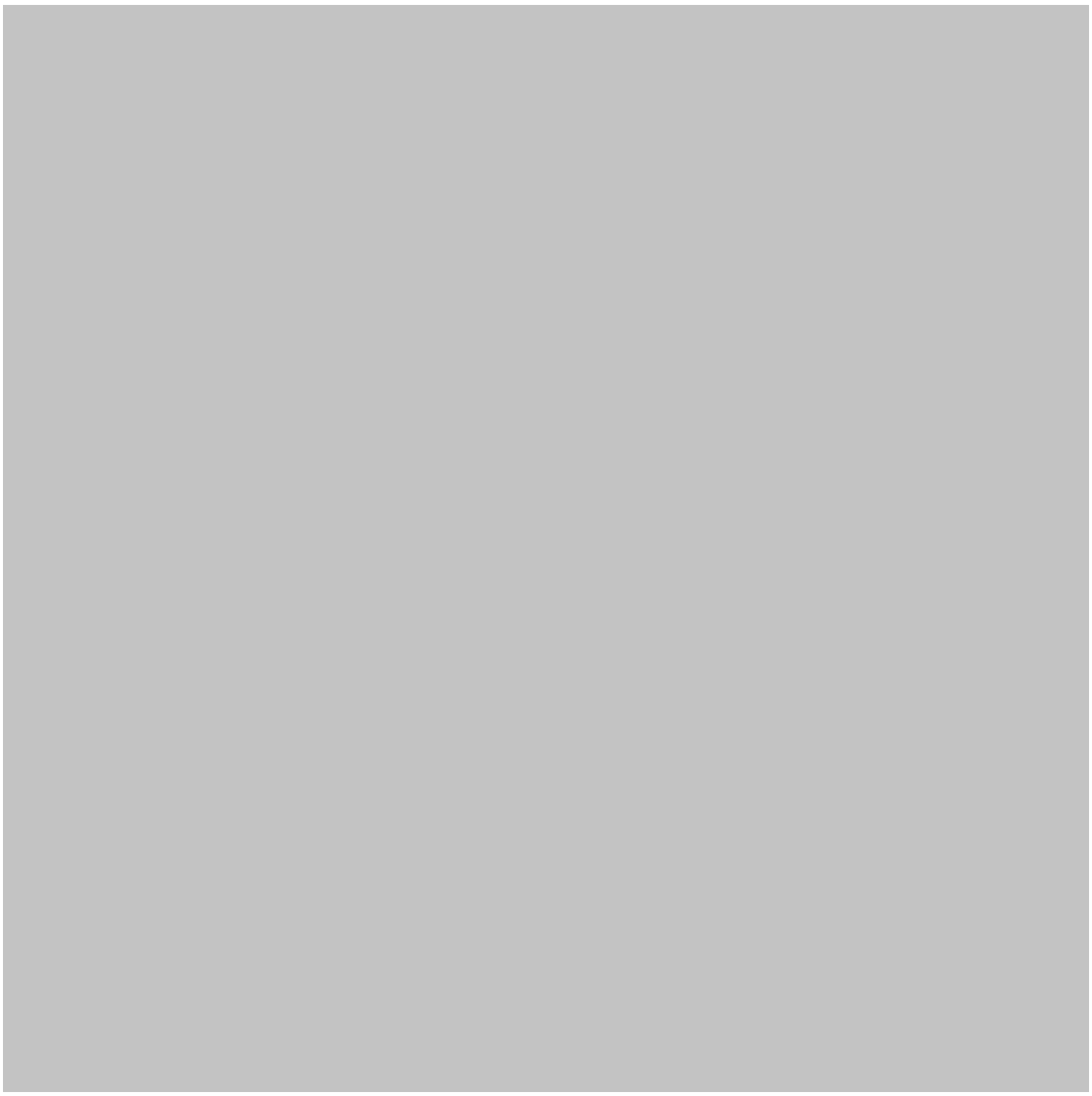
The modified LM317 design which would be allow it to respond to the push button selections can be seen in the following diagram:

The R2 resistors associated need to be calculated with respect to R1 (240 ohms) for setting up the intended push button selected voltage outputs.

High Current LM317 Bench Power Supply

This high current LM317 power supply (https://www.homemade-circuits.com/lm317-with-outboard-current-boost/) can be used universally for any application that requires a high quality regulated high current DC supply, such as car sub woofer amplifiers, battery charges etc.

This power supply is designed to be as versatile as feasible, while also ensuring that the parts count stays low and affordable.



This simple LM317 fixed os adjustable voltage supply satisfies the conditions superbly and is capable of delivering up to 10 amps.

The voltage output is governed by the circuit stage containing R4, R5 and S3; observe that switch S3 is a part of R4.

For getting a fixed voltage output, R4 must be determined for getting zero ohms (fully counter-clockwise). In this situation, switch S3 should be in the open position.

The preset R5 should in that case be tweaked so that the circuit generates a 12 volt output (or anything your personal application requires).

To have an variable output, R4 can be flipped clockwise, with S3 in the closed position, and getting rid of R5 from the circuit.

The output voltage can now be operated by the R4 resistor solely. When the position of SPDT switch S2 is in 1, the highest output current can be accomplished having the two halves of T1 supplying current to the filter stage, in order to increase the overall current output 2 times more.

Having said that, the highest output voltage will be reduced by 50% in this position. It really is a much productive setting considering that the power transistor does not have to drop a significant amount of potential.

In position 2, the maximum voltage practically equals the power specifications of T1. Here, we employed a 24 volt center-tapped transformer for T1.

Lastly, D1 and D2 had been incorporated to safeguard the LM317 IC in case power was switched off with an inductive load at the output

References: http://www.ti.com/lit/ds/symlink/lm317.pdf

https://en.wikipedia.org/wiki/LM317 (https://en.wikipedia.org/wiki/LM317)

You'll also like:

1. Making a Strong RF Discharge Circuit (https://www.homemade-circuits.com/making-strong-rf-discharge-circuit/)
2. What's Ripple Current in Power Supplies (https://www.homemade-circuits.com/whats-ripple-current-in-power-supplies/)
3. 24V to 12V DC Converter Circuit [using Switching Regulator] (https://www.homemade-circuits.com/24-v-to-12-v-dc-converter-circuit/)
4. Making a Regulated 9V Battery Eliminator Circuit (https://www.homemade-circuits.com/making-regulated-9v-battery-eliminator/)
5. Voltage Regulator Circuits and Projects (https://www.homemade-circuits.com/simple-voltage-regulator-circuits-using-transistor-and-zener-diode/)

6. Variable Voltage, Current Power Supply Circuit Using Transistor 2N3055 (https://www.homemade-circuits.com/how-to-make-versatile-variable-voltage/)

About Swagatam

I am an electronics engineer and doing practical hands-on work from more than 15 years now. Building real circuits, testing them and also making PCB layouts by myself. I really love doing all these things like inventing something new, designing electronics and also helping other people like hobby guys who want to make their own cool circuits at home.

And that is the main reason why I started this website homemade-circuits.com, to share different types of circuit ideas..

If you are having any kind of doubt or question related to circuits then just write down your question in the comment box below, I am like always checking, so I guarantee I will reply you for sure!

« How to Configure a Solar Panel System – Living of the Grid (https://www.homemade-circuits.com/how-to-hook-up-wire-grid-tie-solar/)

Automatic Vehicle Headlight Dipper/Dimmer Circuit » (https://www.homemade-circuits.com/how-to-make-automatic-vehicle-headlight/)

Comments

Andrew says
October 26, 2022 at 12:01 am (https://www.homemade-circuits.com/how-to-build-simplest-variable-power/#comment-134117)

I have 4 symmetrical power supplies+42/-42@4.15Amps 300VAC. Need +35/-35 for a parallel LM3886 amplifier x4. Thinking LM317HV/ LM338HV With added transistor for additional current capability. Need small board's and low part count, not a lot of room left, and price concerns. Already over my budget. Please help!!! Can't find a viable solution commercially, don't trust Chinese crap on Ebay. Like to put +/- on 1 board, so just need 4 board's. Want to have board's made, so a Gerber file would be invaluable. Thanks in advance, and for all I've learned from your site, your a 1st class engineer, but a better teacher.

REPLY

Swagatam says
October 26, 2022 at 10:01 am (https://www.homemade-circuits.com/how-to-build-simplest-variable-power/#comment-134125)

You can try the last circuit from the following article:

https://www.homemade-circuits.com/lm317-with-outboard-current-boost/ (https://www.homemade-circuits.com/lm317-with-outboard-current-boost/)

Make sure to add a series resistor with the base of 2N3055. It can be a 1K 1 watt.

REPLY

suat kaleli says
August 14, 2022 at 11:15 am (https://www.homemade-circuits.com/how-to-build-simplest-variable-power/#comment-132339)

Hi Swagatam;
I have made few times circuits all with the parts LM317, BD139 and Tip3055. But unfortunately all were OK at the beginning but then out of order. For instant the latest one at first current control was out of order and then BD139 was out of order and finally the resistance in the circuit was burnt. I need

same purpose circuit again and your advice. Best Regards
P.S.: As known there also are 2 pots 5 K (voltage adjustment) and 10 K (current adjustment) in the circuit. However I have been using 5K plus 1 K for fine adjustment.

REPLY

Swagatam says
August 14, 2022 at 11:57 am (https://www.homemade-circuits.com/how-to-build-simplest-variable-power/#comment-132342)

Hi Suat,
An electronic circuit will go faulty due to some reason, so you will have to find the reason what is causing your circuits to become faulty. It can never go faulty while it is idle, it can become faulty only while it is being used and due to some specific reason such as an overload or over current.
Unless the transistors are over driven they can never become faulty, or may be the transistors are duplicate and not original that could also cause them to burn prematurely.
I can advise if I can see which schematic you are referring to.

REPLY

suat kaleli says
August 15, 2022 at 5:54 pm (https://www.homemade-circuits.com/how-to-build-simplest-variable-power/#comment-132368)

Thanks for the detailed information. The video which I had made the circuit is at the youtube LM317-BD139-TIP3055 (By Zafer Yıldız) contents no schematic. However I can draw it but I do not know how to upload to your site.
But all parts are lm317-bd139-tip3055-10k pot-5 k pot and 220 R.
– 220 R is between the adjustment and out pins of the LM317.
– LM317 output connected to BD139 collector.
– LM317 input and TIP3055 collector are common.(As DC input)
– 10 pot, BD139 emitter and TIP3055 Base are common.
-BD139 base to 10 K pot.
– Other pin of the 10 K pot is common with the pins of LM317 out and BD139 collector.
– Ground to 5K pot and 5 K pot to LM317 adjustment.
– Output voltage is from TIP3055 emitter. Best Regards

REPLY

Swagatam says
August 15, 2022 at 6:24 pm (https://www.homemade-circuits.com/how-to-build-simplest-variable-power/#comment-132369)

If it's possible for you to draw it, then you can upload it to any "free image hosting site" and provide the link to me here, I will check it out.
Or you can refer to the following article and see if your circuit matches with one of the designs presented in the following article:

https://www.homemade-circuits.com/lm317-with-outboard-current-boost/ (https://www.homemade-circuits.com/lm317-with-outboard-current-boost/)

REPLY

suat kaleli says
August 19, 2022 at 1:12 pm (https://www.homemade-circuits.com/how-to-build-simplest-variable-power/#comment-132476)

Hi Swagatam I have uploaded but I am not sure that I made it OK
." alt="LM317 with 2N3055" />

Best regards

REPLY

Swagatam says
August 19, 2022 at 6:54 pm (https://www.homemade-circuits.com/how-to-build-simplest-variable-power/#comment-132490)

Hi Suat,
the circuit looks incorrect to me, especially the BD139 connection.
You don't have to use the BD139, you can connect the LM317 output directly with the base of the 2N3055 through a 1K resistor.

REPLY

suat kaleli says
August 19, 2022 at 10:39 pm (https://www.homemade-circuits.com/how-to-build-simplest-variable-power/#comment-132497)

Thanks for the response Swagatam;
As I wrote before it works I tested and used first circuit about 3 months and then I made the second one both worked but their lifes were limited in shor period. So that means this project works but not stable since all were upset. In the circuit 5K pot control the voltage and 10 K limits the current. As a result my transformer output is AC 24V and it is 100VA (I think that means it is about 4 Amps.) So please advise me a circuit whic is suitable to my transformer and I can also limit the current if required. Best Regards

REPLY

Swagatam says
August 20, 2022 at 9:59 am (https://www.homemade-circuits.com/how-to-build-simplest-variable-power/#comment-132506)

Hi Suat, if the unit was working earlier and then becomes faulty overtime then it is strange. The life of the components can become affected only when there's a fault in the circuit configuration or if the components are not good.

You can modify your design in the following manner and check how long it lasts.
" rel="ugc">

The above circuit will also perfectly work with your 24V 4 amp transformer.
You can also refer to the following posts for more options:

https://www.homemade-circuits.com/how-to-make-versatile-variable-voltage/ (https://www.homemade-circuits.com/how-to-make-versatile-variable-voltage/)
https://www.homemade-circuits.com/how-to-design-a-stabilized-bench-power-supply-circuit/ (https://www.homemade-circuits.com/how-to-design-a-stabilized-bench-power-supply-circuit/)

REPLY

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